Tobias Mwangi-Hartley

PROFESSIONAL SUMMARY

Data Scientist with experience operationalising forecasting and churn propensity through scheduled notebook and batch-scoring workflows, and building experimentation tools using Python, SQL and GCP data tools. Developed a scheduled forecasting workflow in Vertex AI Workbench and weekly batch scoring in BigQuery, with measurable business outcomes. MSc in Statistics with a foundation in modelling uncertainty and experimental design.

TECHNICAL SKILLS

Programming and analysis: Python, SQL, R

Machine learning: LightGBM, XGBoost, feature engineering, SHAP, scikit-learn, statsmodels

Cloud and data platforms: GCP: BigQuery, Vertex AI Workbench; Data workflows: dbt

Applications and testing: Streamlit, Docker, pytest, unit testing

Other: Looker, R Markdown

PROFESSIONAL EXPERIENCE

Data Scientist | Thistlebox *2022–Present*

- Developed and scheduled a weekly LightGBM forecasting workflow in Vertex AI Workbench, producing six-week-ahead recipe-and-region forecasts in BigQuery for operational use in Looker; improved WAPE from 23% to 14% and contributed to an estimated 9% reduction in food waste.
- Built an XGBoost churn propensity model with feature engineering and SHAP analysis; produced weekly batch scores in BigQuery and integrated them into dbt workflows for operational use in retention targeting of top-decile customers.
- Built a Streamlit application for planning and analysing sequential A/B tests, used by approximately 25 people in product and growth; authored the application and Dockerfile, while the data platform team managed deployment to Cloud Run.
- Began writing pytest unit tests for forecasting feature code in 2025; coverage is approximately 60%.
- Delivered the internal "Stats for PMs" training three times.

Junior Data Analyst | Firth Energy Retail *2020–2022*

- Segmented customers and analysed tariffs using R and SQL.
- Automated monthly regulator reports with R Markdown, saving the team approximately two days each month.

EDUCATION

MSc Statistics — Dissertation: Bayesian hierarchical models; coursework/foundation in experimental design and modelling uncertainty.

ADDITIONAL PROJECT

Image Classification Course Project | *2025*

- Fine-tuned a small CNN on a public food-images dataset, achieving approximately 80% accuracy. Learning project completed as part of an online deep learning course.
- Technologies: PyTorch, torchvision, Google Colab.